

Agniveer Vayu Science (Group X) - 22 March 2025 - Memory Based Paper

Physics

Question 1

When the length and area of cross-section both are doubled, then its resistance

Options:

- A. Will become half
- B. Will be doubled
- C. Will remain the same
- D. Will become four times

Answer: C

Solution:

Concept :

Ohm's Law

- If the physical circumstances of the conductor (length, temperature, mechanical strain etc.) remains constant, then the **current** flowing through the conductor is directly proportional to the **potential difference across its two ends** i.e. $I \propto V$
- $V = IR$ or $\frac{V}{I} = R$, where R is a proportionality constant, known as **electric resistance**.

Resistance

- The property of substance by virtue of which it opposes the flow of current through it, is known as the **resistance**.

$$R = \rho \frac{l}{A}$$

Where ρ = resistivity of the material of conductor, l = length of the conductor and A = Area of cross-section of conductor.

Cause of resistance of a conductor –

- It is due to the **collisions of free electrons with the ions or atoms** of the conductor while drifting towards the positive end of the conductor.

Explanation :

Given – Length of first wire (l_1) = L, area of first wire (A_1) = A, length of second wire (l_2) = 2L, area of second wire (A_2) = 2A

The property of substance by virtue of which it opposes the flow of current through it, is known as the **resistance**.

$$R = \rho \frac{l}{A}$$

∴ Resistance of first wire,

$$R_1 = \rho \frac{l_1}{A_1} = \rho \frac{L}{A} \text{ ----- (1)}$$

∴ Resistance of second wire,

$$R_2 = \rho \frac{l_2}{A_2} = \rho \frac{2L}{2A} = \rho \frac{L}{A} \text{ ----- (2)}$$

Divide equation 1 and 2, we get

$$\frac{R_1}{R_2} = \frac{\rho \frac{L}{A}}{\rho \frac{L}{A}} = 1$$

$$R_1 = R_2$$

∴ When the length and area of cross-section both are doubled, then its resistance will remain the same.

Question 2

The surface charge density of a thin spherical shell placed in an air medium is 88.54 c/m^2 The intensity of the electric field measured 12 mm outside the shell from the centre of the shell is $5.625 \times 10^{12} \text{ N/C}$. The thin spherical shell has a radius of:

Options:

A. 10.5 mm

B. 6.0 mm

C. 0.35 mm

D. 9.0 mm

Answer: D

Solution:

Given:

Surface charge density $\sigma = 88.54 \text{ C/m}^2$

Electric field $E = 5.625 \times 10^{12} \text{ N/C}$

Distance measured outside the shell, $x = 12 \text{ mm} = 12 \times 10^{-3} \text{ m}$

Concept:

For a thin uniformly charged spherical shell, the Electric field points outside the shell at a distance x from the centre is

- $E = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{x^2}$
- If the radius of sphere is R, $Q = \sigma 4\pi r^2$
- $E = \frac{1}{4\pi\epsilon_0} \cdot \frac{\sigma \cdot 4\pi R^2}{x^2} = \frac{\sigma}{\epsilon_0} \frac{R^2}{x^2}$

Explanation:

- $E = \frac{\sigma}{\epsilon_0} \frac{R^2}{x^2}$

Put all the given values in above equation to get the value of radius, we get,

$$R^2 = \frac{E \cdot x^2 \cdot \epsilon_0}{\sigma} = \frac{5.625 \times 10^{12} \times 8.85 \times 10^{-12} \times 12 \times 12 \times 10^{-6}}{88.54}$$

$$R^2 = 80.963 \times 10^{-6} \text{ m}$$

$$\Rightarrow R = \sqrt{80.963 \times 10^{-6}} \text{ m} = 8.99 \times 10^{-3} \text{ m} = 9.0 \text{ mm}$$

Hence, the correct answer is $R = 9.0 \text{ mm}$.

Question 3

An electric dipole is formed by two equal and opposite charges q with separation d. The charges have same mass m. It is kept in a uniform electric field E. If it is slightly rotated from its equilibrium orientation, then its angular frequency ω is:

Options:

A. $\sqrt{\frac{qE}{md}}$

B. $\sqrt{\frac{2qE}{md}}$

C. $2\sqrt{\frac{qE}{md}}$

D. $\sqrt{\frac{qE}{2md}}$

Answer: B

Solution:

Concept:

Moment of Inertia:

The formula for moment of inertia is the “sum of the product of mass” of each particle with the “square of its distance from the axis of the rotation”.

The formula of Moment of Inertia is expressed as $I = \sum m_i r_i^2$

Calculation:

The torque is given by the formula:

$$\tau = I\alpha$$

α = Angular acceleration

Now, the torque and electric moment is given by the formula:

$$\tau = -PE \sin \theta$$

For small θ

$$\tau = -PE(\theta) = -qdE\theta \text{ ----(2)}$$

Since, the charge is rotated from equilibrium position.

$$I = \frac{1}{2}mr^2$$

Now, the charge is rotating from its equilibrium position, the moment of inertia is similar to the circular disk.

$$I = \frac{md^2}{2} \text{ ----(3)}$$

On substituting equation (2) and (3) in equation (1),

$$\Rightarrow -qdE\theta = \left(\frac{md^2}{2}\right)\alpha$$

$$\Rightarrow -qE\theta = \left(\frac{md}{2}\right)\alpha$$

$$\Rightarrow -\left(\frac{2qE}{md}\right)\theta = \alpha$$

$$\Rightarrow -\left(\frac{2qE}{md}\right) = \frac{\alpha}{\theta}$$

Angular velocity is vector of angular frequency.

The angular acceleration is change in angular velocity with respect to time.

The angle is the product of angular velocity and time.

$$\Rightarrow -\left(\frac{2qE}{md}\right) = \frac{-\omega^2\theta}{\theta}$$

$$\Rightarrow \left(\frac{2qE}{md}\right) = \omega^2$$

$$\therefore \omega = \sqrt{\frac{2qE}{md}}$$

Question 4

A long solenoid of 50 cm length having 100 turns carries a current of 2.5 A. The magnetic field at the centre of the solenoid is :

$$(\mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1})$$

Options:

A. $6.28 \times 10^{-5}\text{T}$

B. $3.14 \times 10^{-5}\text{T}$

C. $6.28 \times 10^{-4}\text{T}$

D. $3.14 \times 10^{-4}\text{T}$

Answer: C

Solution:

Explanation:

The magnetic field of the center of the solenoid is written as;

$$B = \mu_0 n I \text{ ----(1)}$$

here n is the number of turns per unit length in the solenoid and " I " is the current.

As we know, the number of turns per unit length, $n = N/L$

here, N is the number of turns in the solenoid and L is the length.

Given,

Length = 50 cm, Turns = 100, Current $i = 2.5$ A

Therefore, $n = 100/0.5$ m

$\Rightarrow n = 200$ turns / m

Using equation (1) and putting the values we get;

$$B = \mu_0 n I$$

$$\Rightarrow B = 4\pi \times 10^{-7} \times 200 \times 2.5$$

$$\Rightarrow B = 6.28 \times 10^{-4} \text{ T}$$

Hence, **option 3) is the correct answer.**

Question 5

Which is not an electromagnetic wave?

Options:

- A. Radio
- B. Infrared
- C. Sound
- D. Ultraviolet

Answer: C

Solution:

The correct answer is Sound.

- **Electromagnetic waves or EM waves** are waves that are produced as a result of vibration between an electric field and a magnetic field.
 - In other words, EM waves are composed of oscillating magnetic and electric fields.
 - It includes **radio waves, microwaves, infrared, light, ultraviolet, X-rays, and gamma rays**.
 - All of these waves form part of the electromagnetic spectrum.
 - Electromagnetic radiation consists of electromagnetic waves, which are synchronous oscillations of electric and magnetic fields.
 - Electromagnetic radiation or electromagnetic waves are generated due to the periodic change of electric or magnetic field

Additional Information

Gamma Rays:

- The radioactive decay of atomic nuclei produces **gamma radiation**, a penetrating **electromagnetic field**.
- These electromagnetic waves have the **shortest wavelengths**.
- The wavelength range is **$< 10^{-3}$ nanometres**.

X-ray:

- It is a quick, painless test that produces **images of your bones and other body structures**.
- The density of the material through which X-ray beams pass affects the amount of absorption of the beams.
- The wavelength range is **1 nanometre - 10^{-3} nanometres**.

Infrared ray:

- It is used in **heat-sensitive thermal imaging cameras** and also in the study of human and animal body **heat patterns**.
- **Night vision cameras** which are used in warfare and security also use Infrared rays.
- The wavelength range is **25 micrometers - 2.5 micrometers**.

Ultraviolet rays:

- It is **emitted by the sun** and forms **non-ionizing radiation**. It can also cause **health problems** if we get exposed to it directly.
- Use in industries and medical processes for different purposes like **killing bacteria, creating fluorescent effects**, etc.
- The wavelength range is **400 nanometres - 1 nanometre**

Question 6

The dimensions of energy are:

Options:

A. $[M^1 L^{-1} T^{-1}]$

B. $[M^2 L^2 T^2]$

C. $[M^1L^{-2}T^{-1}]$

D. $[M^1L^2T^{-2}]$

Answer: D

Solution:

The correct answer is option 4): $[M^1L^2T^{-2}]$

Concept:

The dimensional formula of energy is given by,

$$M^1L^2T^{-2}$$

Where,

- M = Mass
- L = Length
- T = Time

$$\text{Energy} = m \times c^2$$

Where m = mass

c = velocity

$$\text{velocity (c)} = \text{Distance} \times [\text{Time}]^{-1} = M^0L^1T^{-1}$$

$$\therefore \text{The dimensional formula of velocity} = M^0L^1T^{-1}$$

$$\text{Energy} = m \times c^2$$

$$\text{Or, } E = [M] \times [L^1T^{-1}]^2 = M^1L^2T^{-2}.$$

Therefore, energy is dimensionally represented as $M^1L^2T^{-2}$.

Question 7

The dimensional formula for specific gravity is given by:

Options:

A. $M^1L^0T^1$

B. $M^1 L^1 T^0$

C. $M^0 L^0 T^0$

D. $M^0 L^1 T^1$

Answer: C

Solution:

Explanation:

- **Specific gravity:** It is also known as **relative density**, it is the **ratio between the density of an object and a reference substance**.
 - The specific gravity can tell us, based on its value, if the object will sink or float in our reference substance.
 - Mathematically it is **Specific gravity = Density of substance / density of water**.
 - **It has no dimension.**
 - Usually, our reference substance is **water** which always has a density of 1 gram per milliliter or 1 gram per cubic centimeter or 1000 kg/m^3 .
- **From the above, it is clear that specific gravity has no dimension. Therefore the dimensional formula of specific gravity is $M^0 L^0 T^0$.**



Additional Information

- **Density:** It is defined as the **mass of a body per unit volume**. We can also say how much mass it can contain in a specific volume.
 - As of dense **material like Iron takes more mass in a given volume than of wood**.
 - Mathematically it gives a formula $\text{Density } (\rho) = \text{mass } (m) / \text{Volume } (V)$
 - SI Unit is kg/m^3 .
 - **$1 \text{ m}^3 = 1000 \text{ liters of water}$.**

Question 8

An object is placed 30 cm before a concave mirror of focal length of 20 cm to get a real image. What will be the distance of the image from the mirror?

Options:

A. 60 cm

B. 30 cm

C. 20 cm

D. 40 cm

Answer: A

Solution:

The correct answer is **60 cm**.



Key Points

- In a spherical mirror, the distance of the object from its pole is called the object distance (u), and the distance of the image from the pole of the mirror is called image distance (v).
- **Focal length (f) is the distance between the principal focus and the pole of the mirror.**
 - **The image distance is given by the formula, $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$**
- In the given question, the object distance is given as 30 cm and the focal length is 20 cm.
 - So, the image distance is, $\frac{1}{v} = \frac{1}{-20} - \frac{1}{-30} = \frac{-3+2}{60} = \frac{-1}{60}$
 - So, the image distance, $v = -60$ cm (real image). Hence option 1 is correct.



Additional Information

- The principal focus is the point on the principal axis at which light rays parallel to the principal axis, after reflection from the mirror.
 - **Concave mirrors are used in torch lights, searchlights, vehicle headlights, shaving mirrors, etc.**
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Question 9

The magnetic moment of a circular coil carrying current is

Options:

- A. Inversely proportional to the length of the square of the wire in the coil
- B. Directly proportional to length of wire
- C. inversely proportional to length of the wire
- D. Directly proportional to the length of the square of the wire in the coil

Answer: D

Solution:

CONCEPT:

- **Magnetic moment (M):** It is defined as the quantity that finds the force which a magnet is able to exert on electric currents and the torque that the magnetic field will exert on it.
 - Magnetic moments can be demonstrated by a lot of things such as a **loop of electric current, an electron, a bar magnet, a molecule, or a planet.**

Formula:

$$M = n i A \dots (\text{for circular loop})$$

where, n = number of turns, i = current flowing through, A = area of cross section.

EXPLANATION:

- As per the above discussion, it's clear that $M \propto n$, $M \propto i$, $M \propto A$.

$$M \propto A \Rightarrow M \propto r^2 \quad (\text{As } l = 2\pi r, i.e. \propto r)$$

Therefore, $M \propto l^2$

Hence the correct option is 4.

Question 10

The cyclic integral of $(\delta Q - \delta W)$ for a process is

Options:

- A. Positive
- B. Negative
- C. Zero
- D. Unpredictable

Answer: C

Solution:

Explanation:

The first law of thermodynamics is a restatement of the law of conservation of energy. It states that energy cannot be created or destroyed in an isolated system; energy can only be transferred or changed from one form to another.

When heat energy is supplied to a gas, two things may occur:

- The internal energy of the gas may change
- The gas may do some external work by expanding

According to the first law of Thermodynamics:

$$\delta Q = \delta W + \Delta U$$

When a process is executed by a system, the change in stored energy of the system is numerically equal to the net heat interaction minus the net work interaction during the process:

$\Delta U = \delta Q - \delta W$ where U is the internal energy which is introduced by this law.

For the cyclic process: $\Delta U = 0$

$$\oint \delta Q = \oint \delta W$$

$$\oint (dQ - dW) = 0$$

The first law of thermodynamics states that for a cyclic process, the cyclic integral of heat is equal to the cyclic integral of work.

Question 11

If the pressure at point A is P and the atmospheric pressure is P_a , then the gauge pressure is equal to:

Options:

- A. P
- B. $P + P_a$
- C. $P - P_a$
- D. None of these

Answer: C

Solution:

CONCEPT:

Atmospheric Pressure:

- The pressure of the atmosphere at any point is equal to the weight of a column of air of unit cross-sectional area extending from that point to the top of the atmosphere.

- At sea level, the atmospheric pressure is given as,

$$\Rightarrow P_a = 1.013 \times 10^5 \text{ Pa} = 1.013 \text{ bar}$$

Absolute Pressure:

- It is the total pressure at a point.

Gauge Pressure:

- Gauge pressure is the difference between absolute pressure and atmospheric pressure.
- Gauge pressure is also known as **relative pressure**.
- The measured pressure is compared to the standard atmospheric pressure at sea level.

$$\Rightarrow P_g = P - P_a$$

Where P_g = gauge pressure, P_a = atmospheric pressure, and P = absolute pressure

EXPLANATION:

- Gauge pressure is the difference between absolute pressure and atmospheric pressure.

$$\Rightarrow P_g = P - P_a$$

Where P_g = gauge pressure, P_a = atmospheric pressure, and P = absolute pressure

- Hence, option 3 is correct.
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Question 12

With two slits spaced 0.2 mm apart, a screen at a distance of 1m, the third bright fringe is found to be displaced 7.5 mm from the central fringe. Find the wavelength of the light used?

Options:

- A. 500 mm
- B. 500 nm
- C. 250 nm
- D. 250 mm

Answer: B

Solution:

Concept:

The path difference is the difference in the distance travelled by two beams when they are scattered in the same direction from different points. It is equal to $d \sin \theta$

Constructive interference (Bright Fringe):

$$d \sin \theta = n\lambda$$

$n = 0$: Central (bright) fringe

$n = 1$: First bright fringe

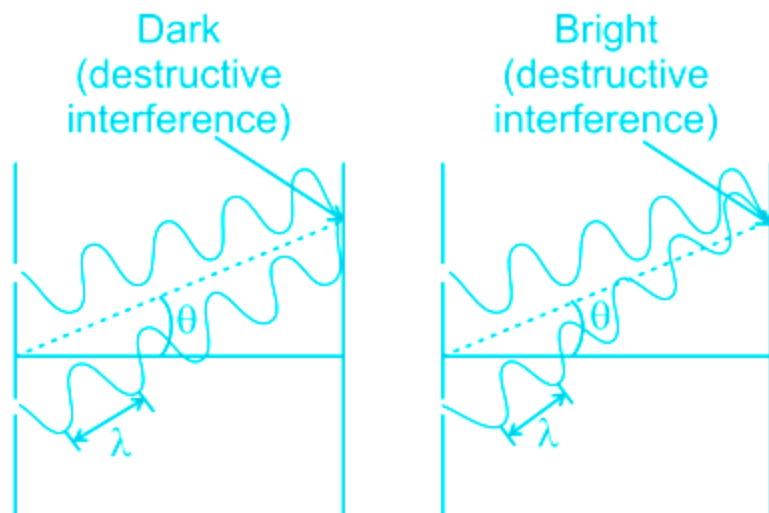
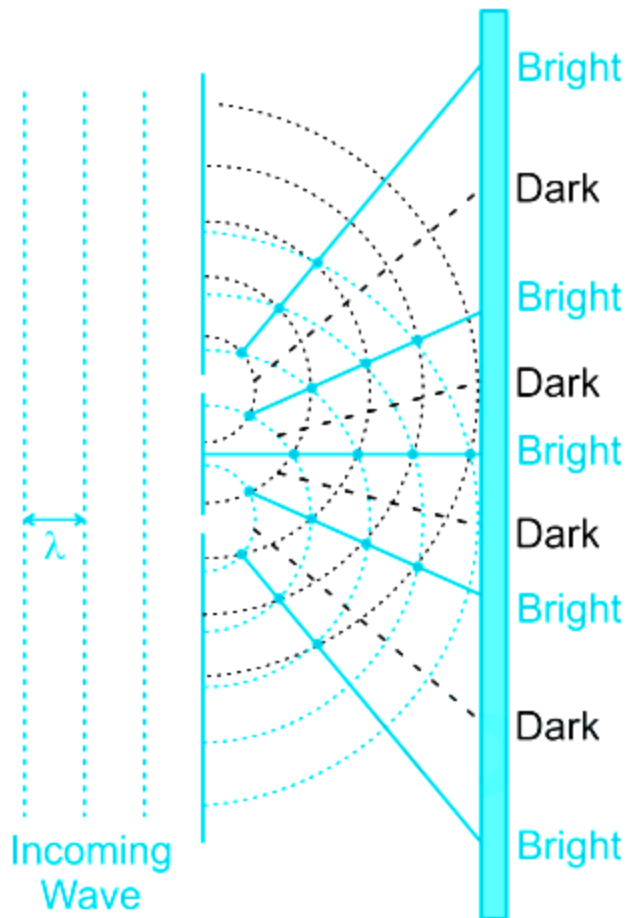
Destructive interference:

$$d \sin \theta = \left(n + \frac{1}{2}\right)\lambda$$

$n = 0$: First dark fringe (next to the central fringe)

$n = 1$: second dark fringe

The separation on the screen between the centres of two consecutive bright fringes or two consecutive dark fringes is called fringe width.



The centres of the bright fringes are obtained at y from central fringe:

$$y = \frac{D}{d} n \lambda$$

The centres of the dark fringes are obtained at y from central fringe:

$$y = \frac{D}{d} \left(n + \frac{1}{2} \right) \lambda$$

The fringe width:

$$w = \frac{D}{d} \lambda$$

D is the separation between the slits and the screen and d is the separation between the slits.

Calculation:

Given: $d = 0.2 \text{ mm}$, $D = 1 \text{ m}$

$$y = \frac{D}{d} n \lambda \Rightarrow 7.5 = \frac{1000}{0.2} \times 3 \lambda \Rightarrow \lambda = 5 \times 10^{-4} \text{ mm} = 500 \text{ nm}$$

Note:

$$1 \text{ m} = 10^3 \text{ mm} = 10^9 \text{ nm}$$

$$1 \text{ mm} = 10^6 \text{ nm}$$

$$10^{-4} \text{ mm} = 10^2 \text{ nm}$$

Question 13

**Kirchoff's current law is based on conservation of _____ while
Kirchoff's voltage law is based on conservation of _____**

Options:

- A. Mass, Charge
- B. Energy, Charge
- C. Charge, Energy
- D. Charge, Mass

Answer: C

Solution:

The correct answer is **Charge, Energy**.



Key Points

- A German physicist named Gustav Kirchhoff created a pair of laws in 1845 that deal with energy and current conservation in electrical circuits.
- These two laws are commonly known as Kirchhoff's Voltage and Current Law.
- Kirchhoff's current law is based on the law of conservation of **charge** while Kirchhoff's voltage law is based on the law of conservation of **energy**.
- Kirchhoff's **Current Law** is also called Kirchhoff's First Law.

- **Kirchhoff's Current Law** states that "The algebraic sum of all currents entering and exiting a node must equal zero."
 - This law describes how a **charge** enters and leaves a wire junction point or node on a wire.
 - Kirchhoff's **Voltage Law** is also called Kirchhoff's Second Law.
 - Kirchhoff's Voltage Law states that the voltage around a loop equals the sum of every voltage drop in the same loop for any closed network and equals zero.
 - Kirchhoff's laws help calculate the electrical resistance of a complex network or impedance in the case of AC and the current flow in different network streams.
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Question 14

n-p-n transistors are preferred over p-n-p transistors because they have

Options:

- A. high mobility of holes
- B. equal to mobility of holes
- C. low mobility of holes
- D. higher mobility of electrons than the mobility of holes in p-n-p transistors

Answer: D

Solution:

Since the **mobility of electrons is higher in NPN transistors, they are preferred more** as it will result in a smooth flow of charge carriers.

Explanation:

Mobility (μ): It is defined as **drift velocity per unit electric field** applied i.e.

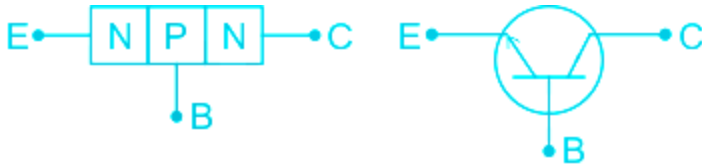
$$\mu = \frac{v_d}{E} = \frac{q\tau}{m}$$

m = effective mass of the charged particle

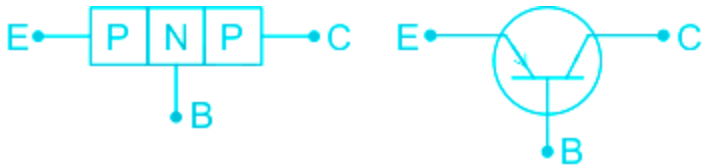
The effective mass of holes is more than that of electrons. This results in less mobility of holes.

NOTES:

NPN transistor: It is formed by sandwiching a thin layer of P-type semiconductor between two N-type semiconductors. Electrons are the carriers responsible for current flow.



PNP transistor: It is formed by sandwiching a thin layer of N-type semiconductor between two P-type semiconductors. Holes are the carriers responsible for current flow.



Question 15

A spring has a spring constant of 1000 N/m and its initial expansion is 10 cm. Calculate the work done in extending the spring from 10 cm to 20 cm.

Options:

- A. 15 J
- B. 20 J
- C. 30 J
- D. 40 J

Answer: A

Solution:

CONCEPT:

- **Elastic potential energy** is the energy stored as a result of applying a force to deform an elastic object.
- The energy is stored until the force is removed and the object **regains its original shape**, doing work in the process.
- This deformation could involve **compressing, stretching, or twisting** the object.
- The elastic potential energy is associated with the work done.
- The **elastic potential energy is equal to the work done (w) in the spring.**

$$\Rightarrow W = \frac{1}{2} kx^2$$

Where k is the spring constant and x is the change in the length of spring from the normal position to the compressed position.

- Many objects are specifically designed to store elastic potential energy inside them,

For example:

1. The coil spring of a wind-up clock.
2. An archer's stretched bow.
3. A stretched slingshot ready to fire.

EXPLANATION:

Given:

$k = 1000 \text{ N/m}$, $x_1 = 10 \text{ cm}$, $x_2 = 20 \text{ cm}$.

- A spring has a spring constant of 1000 N/m and its initial expansion is 10 cm .

Using formula,

$$\Rightarrow W = \frac{1}{2} kx^2$$

Where k is spring constant, x = Displacement of the spring.

$\Rightarrow$ Initially at $x_1 = 10 \text{ cm} = 0.1 \text{ m}$.

$$\Rightarrow W_1 = \frac{1}{2} \times 1000 \times 0.1^2 = 5 \text{ J}$$

at $x_2 = 20 \text{ cm} = 0.2 \text{ m}$.

$$\Rightarrow W_2 = \frac{1}{2} \times 1000 \times 0.2^2 = 20 \text{ J}$$

- Work done to extend the spring up to 20 cm .

$$\Rightarrow W = W_2 - W_1 = 20 - 5 = 15 \text{ J}$$

Question 16

The velocity of sound in air is taken as 332 m/s . If the pressure of a gas is taken double, then what will be the velocity of sound in air?

Options:

A. 996 m/s

B. 664 m/s

C. 166 m/s

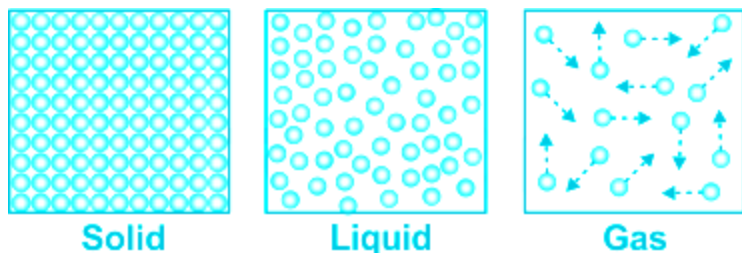
D. 332 m/s

Answer: D

Solution:

CONCEPT:

- As we know in a solid, the particles are **even closer together compared to gas or liquid** and linked by strong covalent bonds so the wave travels **even faster than** it does in either **liquid or air**.



And the **velocity of sound** is given by:

$$v = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma RT}{M}}$$

Hence we can say the speed of sound will be maximum in denser objects, and the velocity of sound changes with density, there are also other factors that can affect the sound's speed, which can be given as

1. Effect of pressure at constant pressure

The velocity of sound is **independent of the pressure of the gas** because as pressure increases, the density of gas also increases. Hence $\frac{P}{\rho}$ ratio remains constant.

$$v = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma RT}{M}} = \text{constant, at constant temperature}$$

2. Effect of temperature

The **velocity of sound increases** with the **rise in temperature**.

$$v = \sqrt{\frac{\gamma RT}{M}} \Rightarrow v \propto \sqrt{T}$$

3. Effect of density

The **velocity of sound decreases**, as **density increases**.

$$v = \sqrt{\frac{\gamma P}{\rho}} \Rightarrow v \propto \frac{1}{\sqrt{\rho}}$$

4. Effect of humidity

- As the **humidity increases**, the density of air decreases which **increases the velocity** of sound.
- Sound travels faster in humid air (rainy season) than in dry air (summer) at the same temperature because:

3. $P_{\text{moist air}} < P_{\text{dry air}} \Rightarrow v_{\text{moist air}} > v_{\text{dry air}}$

Here,

γ = constant for medium.

P & T = pressure and temperature of the medium

ρ & M = density and mass of medium

R = Boltzmann constant

v = velocity/ speed of sound

Explanation:

From the above explanation, we can see that speed of sound will vary with variation in temperature, humidity and elasticity of a medium.

Whereas, The velocity of sound is **independent of the pressure of the gas** because as pressure increases, the density of gas also increases.

Hence the $\frac{P}{\rho}$ ratio will remain constant, because of this speed remains constant.

And thus in this case velocity of sound will remain the same although its pressure becomes double, which means option 4 is correct among all

Question 17

If the mass of a planet is four times the mass of the earth and the radius is half of the earth, then the gravitational acceleration of the planet is:

Options:

A. 16g

B. g/16

C. 4g

D. g

Answer: A

Solution:

CONCEPT:

Acceleration Due to Gravity:

- The force of attraction exerted by the earth on a body is called **gravitational pull or gravity**.
- We know that when a force acts on a body, it produces acceleration. Therefore, a body under the effect of **gravitational pull must accelerate**.
- The acceleration produced in the motion of a body under the effect of gravity is called **acceleration due to gravity**, it is denoted by g .
- The acceleration due to gravity on the surface of the earth is given by,

$$g = \frac{GM}{R^2}$$

Where G = universal gravitational constant, M = mass of the earth, and R = radius of the earth

CALCULATION:

When $M_P = 4M$ and $R_P = \frac{R}{2}$

The gravitational acceleration of the earth is given as,

$$\Rightarrow g = \frac{GM}{R^2} \text{ ----- (1)}$$

The gravitational acceleration of the planet is given as,

$$\Rightarrow g_P = \frac{GM_P}{R_P^2}$$

$$\Rightarrow g_P = \frac{G \times 4M}{\left(\frac{R}{2}\right)^2}$$

$$\Rightarrow g_P = \frac{16 GM}{R^2} \text{ ----- (2)}$$

By equation 1 and equation 2,

$$\Rightarrow g_P = 16g$$

Hence, option 1 is correct.

Question 18

Velocity at mean position of a particle executing S.H.M. is v , then velocity of the particle at a distance equal to half of the amplitude;

Options:

A. $4v$

B. $2v$

C. $\frac{\sqrt{3}}{2}v$

D. $\frac{\sqrt{3}}{4}v$

Answer: C

Solution:

CONCEPT:

Simple harmonic motion

- Simple Harmonic Motion or SHM is defined as a motion in which the **restoring force is directly proportional to the displacement of the body from its mean position.**
 - The **direction of this restoring force is always towards the mean position.**
- **Amplitude:** It is the **maximum displacement** of the particle.
- **Time period:** The **minimum time after which the particle keeps on repeating its motion** is known as the time period.

CALCULATION:

At the mean position,

$$\Rightarrow v = A\omega \text{ -----(1)}$$

- Velocity at half the amplitude is,

$$\Rightarrow v' = \omega \sqrt{A^2 - y^2}$$

For,

$$\Rightarrow y = \frac{A}{2}$$

$$\Rightarrow v' = \omega \sqrt{A^2 - \frac{A^2}{2^2}}$$

$$\Rightarrow v' = \frac{\sqrt{3}}{2} A\omega$$

$$\Rightarrow v' = \frac{\sqrt{3}}{2} v$$

Hence, option 3 is correct.

Question 19

Second law of thermodynamics defines:

Options:

A. Internal Energy

B. Entropy

C. Heat

D. Work

Answer: B

Solution:

Concept:

The **second law of thermodynamics** states that the total entropy of an isolated system always increases over time, or remains constant in ideal cases where the system is in a steady-state or undergoing a reversible process. This increase in entropy accounts for the irreversibility of natural processes. **So the second law gives the entropy term.**

Important Points

- **Zeroth law**– the concept of temperature.
 - **First law**– the concept of internal energy.
 - **Second law**– concept of entropy.
-

Question 20

If wavelength of photon is $6000 \text{ \AA}$, then its energy will be:

Options:

A. 0.66 eV

B. 1.66 eV

C. 2.06 eV

D. 3.5 eV

Answer: C

Solution:

CONCEPT:

Work function (Φ_0):

- The **minimum energy** of incident radiation, required to eject the electrons from the metallic surface is defined as the **work function** of that surface.

$$\Rightarrow \Phi_0 = h\nu_0 = \frac{hc}{\lambda_0}$$

Where h = Planck's constant, c = speed of light, ν_0 = threshold frequency and λ_0 = threshold wavelength.

EXPLANATION:

Given - wavelength of the photon (λ) = 6000 Å

- The threshold wavelength is

$$\Rightarrow \Phi_0 = \frac{hc}{\lambda_0}$$

$$\Rightarrow \Phi_0 = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{6 \times 10^{-7} \times 1.6 \times 10^{-19}} = 2.06 \text{ eV}$$

Question 21

What is the moment of inertia of a ring of mass 'M' and radius 'R' about a tangent to the circle of the ring?

Options:

A. $5MR^2/2$

B. $MR^2/2$

C. $3MR^2/2$

D. $7MR^2/2$

Answer: C

Solution:

CONCEPT:

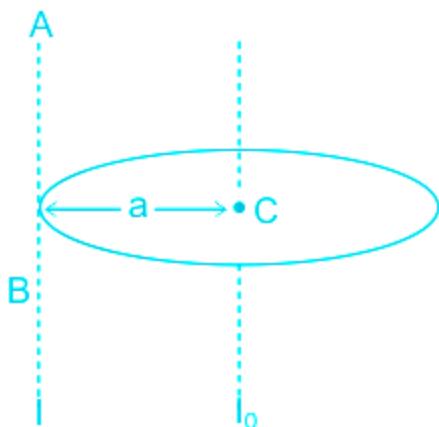
Moment of inertia:

- **Moment of inertia of a rigid body** about a fixed axis is defined as the **sum of the product of the masses of the particles** constituting the body and the **square of their respective distances from the axis of the rotation**.
- Moment of inertia of a particle is

$$I = mr^2$$

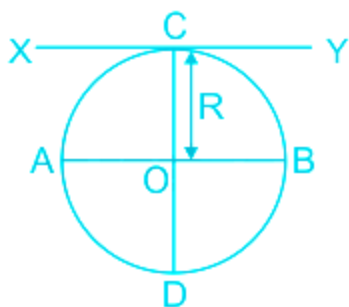
Where r = the perpendicular distance of the particle from the rotational axis.

- **Parallel axis theorem:** Moment of inertia of a body about a given axis I is equal to the **sum of moment of inertia of the body about an axis parallel to the given axis and passing through the centre of mass of the body I_0 and Ma^2** , where 'M' is the mass of the body and 'a' is the perpendicular distance between the two axes.



$$\Rightarrow I_{AB} = I_0 + Ma^2$$

EXPLANATION:



- The **Moment of Inertia of a thin circular ring** of mass 'M' and radius 'R' about a tangent in the plane of the ring is

$$\Rightarrow I_{XY} = I_{AB} + MR^2$$

$$\Rightarrow I_{XY} = \frac{1}{2}MR^2 + MR^2 = \frac{3}{2}MR^2$$



Important Point

Body	Axis of Rotation	Moment of inertia
Uniform circular ring of radius R	perpendicular to its plane and through the center	MR^2
Uniform circular ring of radius R	diameter	$\frac{MR^2}{2}$
Uniform circular disc of radius R	perpendicular to its plane and through the center	$\frac{MR^2}{2}$
Uniform circular disc of radius R	diameter	$\frac{MR^2}{4}$

Solid sphere of radius R	diameter	$\frac{2}{5}MR^2$
A hollow sphere of radius R	diameter	$\frac{2}{3}MR^2$

Question 22

A Carnot engine has an efficiency of $1/6$. When the temperature of the sink is reduced by 62°C , its efficiency is doubled. The temperatures of the source and the sink are, respectively,

Options:

A. 62°C , 124°C

B. 99°C , 37°C

C. 124°C , 62°C

D. 37°C , 99°C

Answer: B

Solution:

Concept:

Carnot engine evaluates that maximum possible efficiency in heat engine during the conversion process of heat and work, it can work between two reservoirs, can possess.

The efficiency of the Carnot engine is given by the formula:

$$\eta = \left(1 - \frac{T_C}{T_H}\right)$$

Where,

T_C = Cold temperature (temperature of sink)

T_H = Hot temperature (temperature of source)

Calculation:

The efficiency before reducing temperature is given as:

$$\Rightarrow \frac{1}{6} = \left(1 - \frac{T_C}{T_H}\right)$$

$$\Rightarrow \frac{T_C}{T_H} = 1 - \frac{1}{6}$$

$$\Rightarrow \frac{T_C}{T_H} = \frac{6-1}{6}$$

$$\therefore \frac{T_C}{T_H} = \frac{5}{6}$$

The efficiency after reducing temperature is given as:

$$\Rightarrow 2 \left(\frac{1}{6} \right) = \left(1 - \frac{T_C - 62}{T_H} \right)$$

$$\Rightarrow \frac{1}{3} = \left(1 - \frac{T_C - 62}{T_H} \right)$$

$$\Rightarrow \frac{1}{3} = \frac{T_H - T_C + 62}{T_H}$$

$$\Rightarrow \frac{1}{3} = 1 - \frac{T_C}{T_H} + \frac{62}{T_H}$$

$$\Rightarrow \frac{1}{3} = 1 - \frac{5}{6} + \frac{62}{T_H}$$

$$\Rightarrow \frac{62}{T_H} = \frac{1}{3} - 1 + \frac{5}{6}$$

$$\Rightarrow \frac{62}{T_H} = \frac{2-6+5}{6}$$

$$\Rightarrow \frac{62}{T_H} = \frac{7-6}{6}$$

$$\Rightarrow \frac{62}{T_H} = \frac{1}{6}$$

$$\therefore T_H = 372 \text{ K}$$

From options, the temperature should be in Kelvin.

$$\Rightarrow T_H = 372 \text{ K} - 273$$

$$\therefore T_H = 99^\circ\text{C}$$

Thus, we have obtained the temperature of the sink.

Now, temperature of the source is:

$$\Rightarrow \frac{T_C}{372} = \frac{5}{6}$$

$$\Rightarrow T_C = \frac{5 \times 372}{6}$$

$$\Rightarrow T_C = 310 \text{ K}$$

$$\therefore T_C = 310 - 273 = 37^\circ\text{C}$$

Question 23

Calculate the work done by a force $\vec{F} = 2\hat{i} + 3\hat{j} + \hat{k}$ on a particle when the particle is displaced by $\vec{S} = \hat{i} + 2\hat{j} - 4\hat{k}$?

Options:

A. 4

B. 6

C. 8

D. 7

Answer: A

Solution:

CONCEPT:

Work done is defined as the dot product of force and displacement.

$$\text{i.e. } W = \text{Force} \cdot \text{Displacement} = \vec{F} \cdot \vec{S}$$

CALCULATION:

Given: A force $\vec{F} = 2\hat{i} + 3\hat{j} + \hat{k}$ displaces a particle by $\vec{S} = \hat{i} + 2\hat{j} - 4\hat{k}$

$$\text{As we know that, } W = \text{Force} \cdot \text{Displacement} = \vec{F} \cdot \vec{S}$$

$$\Rightarrow W = \vec{F} \cdot \vec{S} = (2\hat{i} + 3\hat{j} + \hat{k}) \cdot (\hat{i} + 2\hat{j} - 4\hat{k})$$

$$\Rightarrow W = 2 + 6 - 4 = 4 \text{ units}$$

Hence, **option A** is the correct answer.

Question 24

Which of the following statement is/are correct for an adiabatic process?

Options:

- A. In an adiabatic process the pressure, volume and temperature of the system may change with time.
- B. In an adiabatic process, there is no exchange of heat between the system and surrounding.
- C. An adiabatic process is the part of thermodynamics process.
- D. All of the above

Answer: D

Solution:

CONCEPT:

- **Adiabatic Process:** It is those type of thermodynamics process in which no heat flow between the system and surrounding.
 - In this process pressure, volume, and temperature of the system may change with time.

EXPLANATION:

- The **adiabatic process** comes under the thermodynamics process in which there is **no exchange of heat**.
 - Adiabatic is a **Greek word** means "**heat not passing through**" (a = non, dia = through, bates = heat)
- We can say that $\Delta Q = 0$ (ΔQ is changing in heat, supplied to the system from the surroundings).
- In the **Adiabatic process**, the pressure, volume, and temperature of the system do not remain constant, that is it changing with time.

Question 25

The displacement of a car during its journey is given by $s = xt^2 - yt + z$ where x, y and z are constants and t is the time. The acceleration of the car at $t = 4$ s is

Options:

- A. 4 m/s^2
- B. $2x$
- C. $\sqrt{2} \text{ kmph}$

D. None of the above

Answer: B

Solution:

The correct answer is option 2) i.e. $2x$ units

CONCEPT:

- **Velocity:** The rate of change in displacement is called velocity.
- **Acceleration:** The rate of change in velocity is called acceleration.

Let the displacement of a particle be x .

Velocity, v	$\frac{dx}{dt}$
Acceleration, a	$\frac{dv}{dt}$

CALCULATION:

Given that: Displacement, $s = xt^2 - yt + z$

- The velocity of the car,

$$\Rightarrow v = \frac{ds}{dt} = \frac{d(xt^2 - yt + z)}{dt} = 2xt - y$$

- Acceleration of the car,

$$\Rightarrow a = \frac{dv}{dt} = \frac{d(2xt - y)}{dt} = 2x$$

- The acceleration of the car is time-independent and therefore, at any instant, the acceleration will be $2x$.

Maths

Question 26

Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 5 & 2 & 0 \\ -1 & 6 & 1 \end{bmatrix}$, then the adjoint of A is

Options:

A. $\begin{bmatrix} 2 & -5 & 32 \\ 0 & 1 & -6 \\ 0 & 0 & 2 \end{bmatrix}$

B. $\begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & -6 & 1 \end{bmatrix}$

C. $\begin{bmatrix} -1 & 1 & 0 \\ -5 & -2 & 0 \\ 1 & -6 & -1 \end{bmatrix}$

D. None of these

Answer: D

Solution:

Concept:

Let $A = [a_{ij}]$ be a square matrix.

Steps for obtaining adjoint of the matrix

1. We need to calculate cofactors of all elements

$$\Rightarrow C_{ij} \text{ or } A_{ij} = (-1)^{i+j} M_{ij}$$

Here, M_{ij} = Minor of matrix A

2. Transpose of cofactor is called adjoint of the matrix

$$\Rightarrow \text{Adj}(A) = C^T$$

Calculation:

The adjoint of matrix A is the transpose of the cofactor matrix of A.

$$\text{Cofactor of } 1 = A_{11} = \begin{vmatrix} 2 & 0 \\ 6 & 1 \end{vmatrix} = 2$$

$$\text{Cofactor of } 0 = A_{12} = - \begin{vmatrix} 5 & 0 \\ -1 & 1 \end{vmatrix} = -5$$

$$\text{Cofactor of } 0 = A_{13} = \begin{vmatrix} 5 & 2 \\ -1 & 6 \end{vmatrix} = 32$$

$$\text{Cofactor of } 5 = A_{21} = - \begin{vmatrix} 0 & 0 \\ 6 & 1 \end{vmatrix} = 0$$

$$\text{Cofactor of } 2 = A_{22} = \begin{vmatrix} 1 & 0 \\ -1 & 1 \end{vmatrix} = 1$$

$$\text{Cofactor of } 0 = A_{23} = - \begin{vmatrix} 1 & 0 \\ -1 & 6 \end{vmatrix} = -6$$

$$\text{Cofactor of } -1 = A_{31} = \begin{vmatrix} 0 & 0 \\ 2 & 0 \end{vmatrix} = 0$$

$$\text{Cofactor of } 6 = A_{32} = - \begin{vmatrix} 1 & 0 \\ 5 & 0 \end{vmatrix} = 0$$

$$\text{Cofactor of } 1 = A_{33} = \begin{vmatrix} 1 & 0 \\ 5 & 2 \end{vmatrix} = 2$$

$$\therefore \text{Cofactor matrix} = \begin{bmatrix} 2 & -5 & 32 \\ 0 & 1 & -6 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\text{Adj (A)} = \begin{bmatrix} 2 & 0 & 0 \\ -5 & 1 & 0 \\ 32 & -6 & 2 \end{bmatrix}$$

Question 27

Find $\frac{d^2y}{dx^2}$ if $y = \sin (\log x)$?

Options:

A. $\frac{\sin (\log x) + x \cdot \cos (\log x)}{x^2}$

B. $-\left(\frac{\sin (\log x) + \cos (\log x)}{x^2}\right)$

C. $\frac{x \cdot \sin (\log x) - \cos (\log x)}{2x}$

D. $\frac{\sin (\log x) + \cos (\log x)}{2x}$

Answer: B

Solution:

CONCEPT:

- $\frac{d(\sin x)}{dx} = \cos x$
- $\frac{d(\cos x)}{dx} = -\sin x$
- $\frac{d(\ln x)}{dx} = \frac{1}{x}, \text{ for } x > 0$
- **Quotient Rule:** $\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{g(x) f'(x) - f(x) g'(x)}{[g(x)]^2}$

CALCULATION:

Given: $y = \sin(\log x)$

First let's find out dy/dx

As we know that, $\frac{d(\sin x)}{dx} = \cos x$ and $\frac{d(\ln x)}{dx} = \frac{1}{x}, \text{ for } x > 0$

$$\Rightarrow \frac{dy}{dx} = \frac{\cos(\log x)}{x}$$

Now, again by differentiating the above equation using Quotient rule we get,

As we know that, $\frac{d(\cos x)}{dx} = -\sin x$

$$\Rightarrow \frac{d^2 y}{dx^2} = -\frac{\frac{x \cdot \sin(\log x)}{x} - \cos(\log x)}{x^2}$$

$$\Rightarrow \frac{d^2 y}{dx^2} = -\left(\frac{\sin(\log x) + \cos(\log x)}{x^2} \right)$$

Hence, correct option is 2.

Question 28

Evaluate: $\tan 75^\circ + \cot 75^\circ$

Options:

A. 0

B. ∞

C. 4

D. Undefined

Answer: C

Solution:

Concept:

Some useful formulas are:

$$\cot(90^\circ - \theta) = \tan \theta$$

$$\tan 75^\circ = 2 + \sqrt{3}$$

$$\tan 15^\circ = 2 - \sqrt{3}$$

Calculation:

The given expression is,

$$\tan 75^\circ + \cot 75^\circ$$

$$= \tan 75^\circ + \cot(90^\circ - 15^\circ)$$

$$= \tan 75^\circ + \tan 15^\circ$$

$$= (2 + \sqrt{3}) + (2 - \sqrt{3}) = 4$$

Question 29

Find $\lim_{x \rightarrow 0^+} x \ln x$

Options:

A. 0

B. ∞

C. 1

D. -1

Answer: A

Solution:

We know that the natural logarithm is only defined for positive x and so this is the only limit ($x \rightarrow 0^+$) that makes any sense.

In the limit, we get the indeterminate form i.e. $(0)(-\infty)$. L'Hospital's Rule won't work on products, it only works on quotients. Converting it into a fraction if we rewrite as:

$$\lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x}} = -\frac{\infty}{\infty} \text{ (indeterminate form)}$$

Using L'Hospital's Rule: $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$

$$\lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x}} = \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}} = \lim_{x \rightarrow 0^+} \frac{1}{-\frac{1}{x}} = \lim_{x \rightarrow 0^+} (-x) = 0$$

Question 30

If A is an orthogonal Matrix. Then Inverse of the Matrix $\frac{A^{-1}}{2}$ is

Options:

A. $2(A^{-1})^T$

B. $\frac{(A^T)^{-1}}{2}$

C. $\frac{(A^{-1})^T}{2}$

D. $\frac{(A^T)^{-1}}{4}$

Answer: A

Solution:

Concept:

For the orthogonal matrix,

$$A^T = A^{-1}$$

We know that, $AA^{-1} = I$, So for orthogonal matrix

$$AA^T = I$$

Calculation:

Given:

A is orthogonal Matrix

$$A^T A = A A^T = I$$

$$\text{i.e. } A^{-1} = A^T$$

$$\text{Also } (KA)^n = K^n A^n$$

$$\therefore \left(\frac{A^{-1}}{2}\right)^{-1} = \left(\frac{A^T}{2}\right)^{-1} = 2(A^{-1})^T$$

Question 31

Solve the following:

$$\frac{1}{1+\cos\theta} + \frac{1}{1-\cos\theta} = ?$$

Options:

A. $2\operatorname{cosec}^2\theta$

B. $2\sin^2\theta$

C. $2\sec^2\theta$

D. $2\tan^2\theta$

Answer: A

Solution:

Given:

$$\frac{1}{1+\cos\theta} + \frac{1}{1-\cos\theta} = ?$$

Formula used:

$$\frac{1}{1+\cos\theta} + \frac{1}{1-\cos\theta} = \frac{(1-\cos\theta) + (1+\cos\theta)}{(1+\cos\theta)(1-\cos\theta)}$$

Calculations:

$$\begin{aligned} & \frac{1}{1+\cos\theta} + \frac{1}{1-\cos\theta} \\ \Rightarrow & \frac{(1-\cos\theta) + (1+\cos\theta)}{(1+\cos\theta)(1-\cos\theta)} \\ \Rightarrow & \frac{1-\cos\theta + 1+\cos\theta}{1-\cos^2\theta} \\ \Rightarrow & \frac{2}{\sin^2\theta} \end{aligned}$$

$$\Rightarrow \frac{2}{\sin^2 \theta} = 2 \operatorname{cosec}^2 \theta$$

∴ The correct answer is option (1).

Question 32

In a class of 15 students, 5 fail in a test. Marks of remaining 10 students are 9, 6, 8, 7, 8, 9, 5, 6, 7 and 4. The median of marks of all 15 students is:

Options:

A. 5

B. 6

C. 7

D. 4

Answer: B

Solution:

The correct answer is **6**.

Key Points

Five students failed the test. Their scores cannot be derived. But we can assume that one or more students might get the same marks.

Let's denote the test scores received by the failed students as F, where $F = 0, 1, 2, 3$. As the score of the students who passed the test starts from 4, F must be less than 4.

Students who passed the test have scored 9, 6, 8, 7, 8, 9, 5, 6, 7 and 4.

Arranging their scores in ascending order, we have: F, F, F, F, F, 4, 5, 6, 6, 7, 7, 8, 8, 9, 9

n = total number of students = 15, an odd number.

The median in the series would be $\frac{(n+1)}{2}$ th term = $(15 + 1) / 2 = 16 / 2 = 8$ th number.

The 8th number in the series is 6.

Question 33

Find the angle θ between the vectors $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{b} = 3\hat{i} - 2\hat{j} + \hat{k}$?

Options:

A. $\cos^{-1} \left(\frac{4}{7} \right)$

B. $\cos^{-1} \left(\frac{5}{7} \right)$

C. $\cos^{-1} \left(\frac{5}{9} \right)$

D. None of these

Answer: B

Solution:

Concept:

For any two vectors, $\vec{a}$ and $\vec{b}$ we have $\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| \times |\vec{b}|}$

Calculation:

Here, we have two vectors $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{b} = 3\hat{i} - 2\hat{j} + \hat{k}$.

$$\Rightarrow |\vec{a}| = \sqrt{1^2 + (-2)^2 + 3^2} = \sqrt{14} \text{ and } |\vec{b}| = \sqrt{3^2 + (-2)^2 + 1^2} = \sqrt{14}$$

$$\Rightarrow \vec{a} \cdot \vec{b} = (\hat{i} - 2\hat{j} + 3\hat{k}) \cdot (3\hat{i} - 2\hat{j} + \hat{k}) = 3 + 4 + 3 = 10$$

By, substituting the values of $|\vec{a}|$, $|\vec{b}|$ and $\vec{a} \cdot \vec{b}$ in $\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| \times |\vec{b}|}$, we get

$$\Rightarrow \cos \theta = \frac{10}{\sqrt{14} \times \sqrt{14}} = \frac{5}{7}$$

$$\Rightarrow \theta = \cos^{-1} \left(\frac{5}{7} \right)$$

Hence, **option B** is the correct answer.

Question 34

What is the probability that a leap year selected at random contains 53 Sundays?

Options:

A. $1 / 7$

B. $2 / 7$

C. $7 / 366$

D. $26 / 183$

Answer: B

Solution:

Concept:

Total no. of days in a leap year = 366

Total no. of days in a week = 7

Total no. of weeks = $366/7 = 52$ weeks and 2 days

Calculation:

To find: Probability that a leap year selected at random will contain 53 Mondays

As we know, in a leap year there will be fifty-two weeks and two days.

These 2 days can be:

(Sun, Mon), (Mon, Tue), (Tue, Wed), (Wed, Thu), (Thu, Fri), (Fri, Sat) and (Sat, Sun)

Total outcomes = 7

Favourable outcomes = (Sun, Mon), (Sat, Sun) = 2

Hence the probability of getting 53 Sundays in a leap year = $2/7$

Question 35

Find the number of permutations of 4 letters taken from the word EXAMINATION.

Options:

A. 1504

B. 2520

C. 2552

D. 2454

Answer: D

Solution:

Calculation:

Given word = EXAMINATION

Total number of single letters = 8 (E, X, A, M, I, A, T, O)

Duplicate letters = 3 (AA, II, NN)

∴ Possible ways of selecting 4 letters are :

1. 4 from single = ${}^8C_4 \times 4! = 1680$

2. 2 from single and 1 pair of the duplicated letters = ${}^3C_1 \times {}^7C_2 \times \frac{4!}{2!} = 756$

3. 2 pair of duplicated letters = ${}^3C_2 \times \frac{4!}{2!2!} = 18$

Total number of ways = 2454

∴ The number of permutation of 4 letters taken from the word EXAMINATION is 2454.

The correct answer is Option 4.

Question 36

If $y = e^{5x}$ then $\frac{d^2y}{dx^2} =$

Options:

A. 25y

B. 15y

C. 10y

D. $5y$

Answer: A

Solution:

Concept:

Second-order Derivative.

$$\frac{d^2f(x)}{dx^2} = \frac{d}{dx} \left[\frac{d}{dx} f(x) \right]$$

$$\frac{d}{dx} e^{f(x)} = e^{f(x)} f'(x)$$

Calculations:

$$y = e^{5x}$$

Differentiating w.r. to x on both side, we get

$$\Rightarrow \frac{dy}{dx} = 5e^{5x}$$

Again Differentiating w.r. to x on both sides, we get

$$\Rightarrow \frac{d^2y}{dx^2} = 5 \times 5e^{5x}$$

$$\Rightarrow \frac{d^2y}{dx^2} = 25y$$

Question 37

$$\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2} \text{ is}$$

Options:

A. 2

B. 0

C. 1

D. 4

Answer: A

Solution:

Concept:

- **Indeterminate Forms:** Any expression whose value cannot be defined, like $\frac{0}{0}, \pm \frac{\infty}{\infty}, 0^0, \infty^0$ etc.
- For the indeterminate form $\frac{0}{0}$, first try to rationalize by multiplying with the conjugate, or simplify by cancelling some terms in the numerator and denominator. Else, use the L'Hospital's rule.
- **L'Hospital's Rule:** For the differentiable functions $f(x)$ and $g(x)$, the $\lim_{x \rightarrow c} \frac{f(x)}{g(x)}$, if $f(x)$ and $g(x)$ are both 0 or $\pm\infty$ (i.e. an Indeterminate Form) is equal to $\lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$ if it exists.
- $\frac{d}{dx}(\cos x) = -\sin x$

Calculations:

Consider $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$

$= \frac{0}{0}$ which is an indeterminate form.

Using the L'Hospital rule, we have

$$= \lim_{x \rightarrow 0} \frac{-(-\sin 2x) \times (2)}{2x}$$

$$= \lim_{x \rightarrow 0} \frac{\sin 2x}{x} \quad |\because \sin 0 = 0|$$

$= \frac{0}{0}$ which is again an indeterminate form.

Again using the L'Hospital rule, we have

$$= \lim_{x \rightarrow 0} \frac{2 \cos 2x}{1}$$

Applying limit as $x \rightarrow 0$ we get

$$= 2$$

$$\therefore \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2} = 2$$

Hence, the correct answer is option 1).

Question 38

Find the modulus of $z = (1 - i)^4$?

Options:

- A. 2
- B. 4
- C. - 4
- D. None of these

Answer: B

Solution:

CONCEPT:

- $i^2 = -1$
- If $z = x + iy$ then $|z| = \sqrt{x^2 + y^2}$

CALCULATION:

Given: $z = (1 - i)^4$ First let's simplify the expression $(1 - i)^4$

$$\Rightarrow (1 - i)^2 = 1 + i^2 - 2i$$

As we know that, $i^2 = -1$

$$\Rightarrow (1 - i)^2 = -2i$$

Since $(1 - i)^4 = (1 - i)^2 \times (1 - i)^2$ we get:

$$\Rightarrow (1 - i)^4 = (-2i)^2 = -4$$

$$\Rightarrow z = -4 + 0i$$

As we know that, if $z = x + iy$ then $|z| = \sqrt{x^2 + y^2}$

Here, $x = -4$ and $y = 0$

$$\Rightarrow |z| = \sqrt{(-4)^2 + 0^2} = \pm 4$$

As we know that, $|z|$ denotes the distance between origin and z in the argand plane. So, $|z|$ cannot be negative

$$\Rightarrow |z| = 4$$

Hence, correct option is 2.

Question 39

Differentiate $\cos(\cos(x))$ with respect to x ?

Options:

A. $\sin(\sin(x))(\sin(x))$

B. $\sin(\sin(x))(\cos(x))$

C. $\cos(\sin(x))(\sin(x))$

D. $\sin(\cos(x))(\sin(x))$

Answer: D

Solution:

Concept:

Chain rule:

Let $y = f(v)$ be a differentiable function of v and $v = g(x)$ be a differentiable function of x then $\frac{dy}{dx} = \frac{dy}{dv} \cdot \frac{dv}{dx}$

Calculation:

Let $y = \cos(\cos(x))$

$$\Rightarrow \frac{dy}{dx} = \frac{d}{dx} \cos(\cos(x))$$

$$\Rightarrow \frac{dy}{dx} = -\sin(\cos(x)) \frac{d}{dx} (\cos(x))$$

$$\Rightarrow \frac{dy}{dx} = -\sin(\cos(x))(-\sin(x))$$

$$\Rightarrow \frac{dy}{dx} = \sin(\cos(x))(\sin(x))$$

Question 40

The arithmetic mean of the squares of the first n natural numbers is

Options:

A. $\frac{n(n+1)(2n+1)}{6}$

B.

$$\frac{n(n+1)(2n+1)}{2}$$

C.

$$\frac{(n+1)(2n+1)}{6}$$

D. $\frac{(n+1)(2n+1)}{3}$

Answer: C

Solution:

Concept:

Sum of the first n natural numbers is given by $\frac{n(n+1)}{2}$

Sum of the squares of first n natural numbers is given by $\frac{n(n+1)(2n+1)}{6}$

Arithmetic mean: The arithmetic mean is the sum of all the numbers in a data set divided by the quantity of numbers in that set.

Calculation:

We have to find the arithmetic mean of the squares of the first n natural numbers,

As we know **Sum of the squares of first n natural numbers** is given by $\frac{n(n+1)(2n+1)}{6}$

Now,

Arithmetic mean =

$$\frac{\text{Sum of the squares of first n natural numbers}}{n}$$

$$= \frac{\frac{n(n+1)(2n+1)}{6}}{n}$$

$$= \frac{(n+1)(2n+1)}{6}$$

Question 41

The integral $\int_0^{\pi/2} |\sin x - \cos x| dx$ is equal to

Options:

A. $2\sqrt{2}$

B. $2(\sqrt{2} - 1)$

C. $2(\sqrt{2} + 1)$

D. $\sqrt{2} + 1$

Answer: B

Solution:

Given:

$$\int_0^{\pi/2} |\sin x - \cos x| dx$$

Calculation:

$$\text{Let } I = \int_0^{\pi/2} |\sin x - \cos x| dx$$

$$\Rightarrow I = - \int_0^{\pi/4} (\sin x - \cos x) dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x) dx$$

$$\Rightarrow I = - \left[-\cos x - \sin x \right]_0^{\pi/4} + \left[-\cos x - \sin x \right]_{\pi/4}^{\pi/2}$$

$$\Rightarrow I = - \left[-\cos \frac{\pi}{4} - \sin \frac{\pi}{4} - (-\cos 0 - \sin 0) \right] + \left[-\cos \frac{\pi}{2} - \sin \frac{\pi}{2} - \left(-\cos \frac{\pi}{4} - \sin \frac{\pi}{4} \right) \right]$$

$$\Rightarrow I = - \left[-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} - (-1 - 0) \right] + \left[0 - 1 - \left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \right) \right]$$

$$\Rightarrow I = -(-\sqrt{2} + 1) + (-1 + \sqrt{2})$$

$$\Rightarrow I = (\sqrt{2} - 1) + (\sqrt{2} - 1)$$

$$\Rightarrow I = 2(\sqrt{2} - 1)$$

$\therefore$ The value of the integral is $2(\sqrt{2} - 1)$

Question 42

Simplify the following: $(2 + \sqrt{y})^4 + (2 - \sqrt{y})^4$

Options:

A. $y^2 + 24y + 16$

B. $2y^2 + 48y + 32$

C. $y^2 + 48y + 32$

D. $2y^2 + 24y + 32$

Answer: B

Solution:

Formula Used:

$$(1 + x)^n = C_0 + C_1 \cdot x + C_2 \cdot x^2 + \dots + C_r \cdot x^r + \dots + C_n \cdot x^n$$

Calculation:

By using the above relation

$$(x + a)^4 = x^4 + {}^4C_1 \cdot x^3 \cdot a + {}^4C_2 \cdot x^2 \cdot a^2 + {}^4C_3 \cdot x \cdot a^3 + {}^4C_4 \cdot x^0 \cdot a^4$$

Similarly,

$$(x - a)^4 = x^4 - {}^4C_1 \cdot x^3 \cdot a + {}^4C_2 \cdot x^2 \cdot a^2 - {}^4C_3 \cdot x \cdot a^3 + {}^4C_4 \cdot x^0 \cdot a^4$$

Adding both of relations

$$(x + a)^4 + (x - a)^4 = 2 [x^4 + 6 \cdot x^2 \cdot a^2 + a^4]$$

Put $x = 2$ & $a = \sqrt{y}$

$$\Rightarrow (2 + \sqrt{y})^4 + (2 - \sqrt{y})^4$$

$$\Rightarrow 2 \times [(2)^4 + 6 \cdot (2)^2 \sqrt{y}^2 + \sqrt{y}^4]$$

$$\Rightarrow 2 \times (16 + 6 \times 4y + y^2)$$

$$\therefore 2y^2 + 48y + 32$$

Question 43

If $A = \{1, 2, 3, 4, 5\}$, $B = \{1, 4, 5\}$ and $R = \{(a, b): a \in A, b \in B \text{ and } a < b\}$ then find the domain of R ?

Options:

- A. $\{1, 2, 3, 4\}$
- B. $\{1, 2, 3, 4, 5\}$
- C. $\{1, 3, 4, 5\}$
- D. None of these

Answer: A

Solution:

Concept:

Domain of a Relation: Let R be a relation from set A to set B . Then, the set of all first components of the ordered pair belonging to relation R forms the domain of the relation R .

i.e $\text{Domain}(R) = \{a: (a, b) \in R\}$

Calculation:

Given: $A = \{1, 2, 3, 4, 5\}$, $B = \{1, 4, 5\}$ and $R = \{(a, b): a \in A, b \in B \text{ and } a < b\}$

Here, we have to find the domain of R .

$\Rightarrow R = \{(1, 4), (1, 5), (2, 4), (2, 5), (3, 4), (3, 5), (4, 5)\}$

As we know that, $\text{Domain}(R) = \{a: (a, b) \in R\}$

So, for the given relation R , we have

$\text{Domain}(R) = \{1, 2, 3, 4\}$

Question 44

Find the intercepts cut off by the plane $x + 2y - 4z = 8$.

Options:

- A. (1, 2, 4)
- B. (1, 2, -4)
- C. (8, 4, 2)
- D. (8, 4, -2)

Answer: D

Solution:

Concept:

The **intercept form of the plane** is given by: $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

Where a is the x-intercept and b is the y-intercept and c is the z-intercept.

Calculation:

Given: $x + 2y - 4z = 8$

$$\Rightarrow \frac{x}{8} + \frac{2y}{8} - \frac{4z}{8} = 1$$

$$\Rightarrow \frac{x}{8} + \frac{y}{4} + \frac{z}{-2} = 1$$

As we know that intercept form of the plane is given by: $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

Hence, the intercepts cut off by the given plane are $(a, b, c) = (8, 4, -2)$

Question 45

The number of terms in the expansion of $(x + y + z)^{10}$ is

Options:

- A. 11
- B. 33
- C. 66
- D. None of these

Answer: C

Solution:

Concept:

- Number of terms in the expansion of $(x_1 + x_2 + x_3 + \dots + x_r)^n$ is $= {}^{n+r-1}C_n$, Where n = exponent of the term to be expanded, r = number of terms to be expanded
- ${}^nC_r = \frac{n!}{(n-r)! r!}$

Explanation:

Number of terms in $(x + y + z)^{10} = {}^{10+3-1}C_{10}$ (Since, n = 10 and r = 3)

$$\Rightarrow {}^{12}C_{10}$$

$$\text{Since, } {}^{12}C_{10} = \frac{12!}{(12-10)! 10!}$$

$$\Rightarrow {}^{12}C_{10} = 66$$

Question 46

The area bounded by the curve $y = \sin x$ between $x = 0$ and $x = \pi$ is

Options:

A. 0

B. 1

C. 2

D. 4

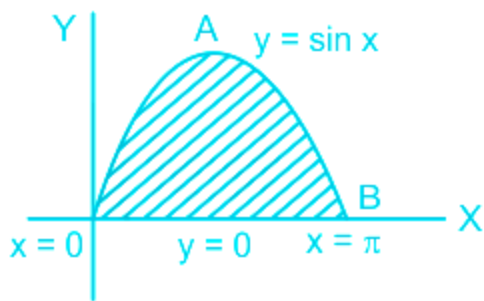
Answer: C

Solution:

Given:

Curve $y = \sin x$

Area bounded between $x = 0$ and $x = \pi$



Calculation:

We have,

$$\Rightarrow A = \int_0^{\pi} \sin x \, dx$$

$$\Rightarrow A = [-\cos x]_0^{\pi}$$

$$\Rightarrow A = -[\cos \pi - \cos 0]$$

$$\Rightarrow A = -[(-1) - 1]$$

$$\Rightarrow A = 2$$

$\therefore$ The area bounded by the curve is 2

Question 47

The axis of symmetry of the parabola $y^2 = x$ is:

Options:

A. Y axis

B. X axis

C. -Y axis

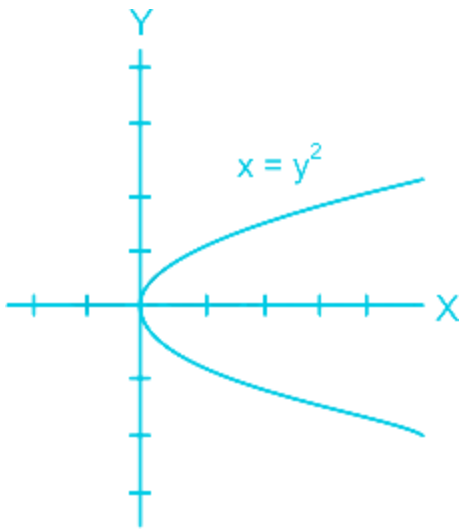
D. -X axis

Answer: B

Solution:

Explanation:

$y^2 = x$ can be drawn as



∴ The axis symmetry of the parabola is the X-axis.

Question 48

If $A = \begin{bmatrix} 1 & -11 \\ 2 & -10 \\ 1 & 0 & 0 \end{bmatrix}$ then $A^5 =$

Options:

- A. A
- B. Identity Matrix
- C. Null Matrix
- D. A^{-1}

Answer: D

Solution:

Calculation:

Given $A = \begin{bmatrix} 1 & -11 \\ 2 & -10 \\ 1 & 0 & 0 \end{bmatrix}$

$$A^2 = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 2 \\ 1 & -1 & 1 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I \text{ (identity Matrix)}$$

$$A^5 = A^2 \cdot A^3 = A^2 \cdot I$$

$$A^5 = A^2 (A A^{-1}) = A^3 A^{-1} = I \cdot A^{-1}$$

$$\text{Therefore, } A^5 = A^{-1}$$

The correct answer is option (4).

Question 49

Evaluate:- $\int \sin x \sec^2 x \, dx$

Options:

A. $\sin x + c$

B. $\tan x + c$

C. $\cot x + c$

D. $\sec x + c$

Answer: D

Solution:

Concept:

Relation and formula used:

$$\sec x = \frac{1}{\cos x}$$

$$\tan x = \frac{\sin x}{\cos x}$$

$$\int \sec x \tan x \, dx = \sec x$$

Calculation:

$$\text{Let, } I = \int \sec^2 x \sin x \, dx$$

$$\Rightarrow I = \int \sec x \frac{1}{\cos x} \sin x \, dx$$

$$\Rightarrow I = \int \sec x \tan x \, dx$$

$$\Rightarrow I = \sec x + c$$

$$\therefore \text{The value of } \int \sec^2 x \sin x \, dx = \sec x + c$$

Question 50

If the median of the observations 2, 3, 5, 6, x, 8, 9, is 6 then x CANNOT be equal to:

Options:

- A. 8
- B. 1
- C. 10
- D. 7

Answer: B

Solution:

The correct answer is 1.

Key Points

Given the observations 2, 3, 5, 6, x, 8, 9, to determine the median, we arrange the observations in ascending order: 2, 3, 5, 6, x, 8, 9.

Since the number of observations is odd, the median is the middle value in the ordered list. In this case, the fourth observation, which is 6, represents the median.

- To identify the values that x cannot be equal to, we need to consider the possible positions of x in the ordered list.
 - If x is less than 6, it would be positioned to the left of 6 in the list. In such a case, the median would be greater than 6, as the middle value would shift to a higher number. Therefore, x cannot be less than 6.

- If **x is greater than 6**, it would be positioned to the right of 6 in the ordered list. In this case, the median would still be 6 since it represents the middle value. Therefore, x can be greater than 6.
- If **x equals 6**, it would coincide with the median, as it represents the middle value in the ordered list. Therefore, x can be equal to 6.

Hence, considering the given information that the median is known to be 6, it can be concluded that x cannot be equal to 1, as it is less than 6.

English

Question 51

Change the given statement to indirect speech.

Tom: "I'll be there at eight. Is that all right?"

Options:

- A. Tom said that he would be there at eight and he checked if it was all right.
- B. Tom said that he will be there at eight and asked if it was all right.
- C. Tom told that he would be there at eight. And asked if it is all right.
- D. Tom said he will there at eight and asked if that is all right.

Answer: A

Solution:

The correct answer is '**Tom said that he would be there at eight and he checked if it was all right**'.



Key Points

- While changing the given sentence from **direct to indirect**, we need to follow the given steps -
- The sentence is in **simple future tense**, so here '**will**' changes to '**would**' in indirect speech.
- The pronoun will change according to the subject of the reporting verb. So, here '**I**' to '**he**'. (the verb is not mentioned in the given sentence)
- We will use the reporting verb '**said**' followed by '**that**'.
- We will use '**and**' to connect the two sentences within the reported speech.
- Since the second sentence in the reported speech starts with **an interrogative verb**, we will use 'if' in indirect speech.
- Thus, based on the given points, the correct answer is **Option 1**.

Correct sentence is indirect speech - *Tom said that he would be there at eight and he checked if it was all right.*

Question 52

A sentence has been given with a blank to be filled with an appropriate option. Choose the correct alternative.

John rushed to the spot and saw his wife and kid _____ on the road.

Options:

A. lain

B. lying

C. lie

D. laying

Answer: B

Solution:

The correct answer is 'Option 2 i.e. (lying)'



Key Points

- The correct verb to use in this context is "lying" as it refers to being in or assuming a horizontal or resting position on a surface.
- "Lying" is the present participle of the verb "to lie."
- Therefore, the sentence should read, "John rushed to the spot and saw his wife and kid lying on the road."
- This makes Option 2 (lying) the correct choice to fill the blank.

Therefore, the correct sentence is: "John rushed to the spot and saw his wife and kid lying on the road."



Additional Information

- "lain" - This is the past participle form of the verb "to lie" but is not suitable in this context.
 - "lie" - This is the base form of the verb, not correct in this sentence's tense structure.
 - "laying" - This form of the verb typically requires an object, e.g., 'laying the blanket on the bed', and is not correct here.
-

Question 53

Choose the part of the sentence that is incorrect in questions.

He said (1)/ that the horse (2)/ died at (3)/ the night.(4)

Options:

A. (1)

B. (2)

C. (3)

D. (4)

Answer: C

Solution:

The correct answer is '3 - died at.'



Key Points

- The phrase "**died at the night**" is incorrect. The correct preposition to use with "night" is "**in**," not "at."
- "At" can be used with specific times, like "**at 10 o'clock**," but with periods of the day like "night," the correct preposition is "**in**."
- Therefore, the correct sentence should read: "**He said that the horse died in the night.**"
- The rest of the sentence is grammatically correct and follows the proper tense and structure for indirect speech.

Therefore, the correct answer is Option 3.

Question 54

In the following questions, out of the four alternatives choose the one which can be substituted for the given words / sentence.

'Edible' means that which is:

Options:

A. Worth eating

B. Can not be deleted

C. Can be edited

D. Edifying

Answer: A

Solution:

The correct answer is option 1 i.e. 'Worth eating'.



Key Points

- **Edible:** (खाद्य) fit to be eaten.
 - **Example:** "The fruits and vegetables in the market were all edible."
- **Worth eating:** This is a synonym for edible.

Therefore, the correct answer is - **Worth eating**



Additional Information

- **Can not be deleted:** This refers to something that cannot be removed or erased.
 - **Can be edited:** This refers to something that can be changed or modified.
 - **Edifying:** This refers to something that is morally or intellectually uplifting.
-

Question 55

Select the word **OPPOSITE** in meaning to the given word.

Deficient

Options:

A. Skimpy

B. Broken

C. Ample

D. Terrible

Answer: C

Solution:

The correct answer is: **Option 3.**

Key Points

- The word "**Deficient**" means lacking in some necessary quality or element; not enough in amount or degree. (अपर्याप्त)
 - **Example:** The diet was deficient in vitamins, leading to health problems.
- "**Ample**" refers to being more than enough in quantity, size, or scope; fully sufficient or more than adequate. (प्रचुर)
 - **Example:** There was ample space in the room for all the guests to sit comfortably.
- Hence, we can infer that the opposite of '**Deficient**' is '**Ample**'.

Therefore, the correct answer is: **Option 3.**

Additional Information

Here are the other options explained along with their Hindi meanings and example sentences:

- **Skimpy** (अल्प): Lacking in size, fullness, or quantity; meager.
 - **Example:** The portions of food served were quite skimpy, leaving everyone still hungry.
 - **Broken** (टूटा हुआ): Separated into parts or pieces due to being damaged.
 - **Example:** The broken vase lay in fragments on the floor.
 - **Terrible** (भयानक): Extremely or distressingly bad or serious.
 - **Example:** The weather was so terrible that we decided to stay indoors all day.
-

Question 56

Find the part of the given sentence that has an error in it. If there is no error, choose 'No error'.

It is believed that difficult tasks demand lot of our resources.

Options:

- A. It is believed that
- B. No error
- C. difficult tasks demand lot
- D. of our resources.

Answer: C

Solution:

The correct answer is 'Option 3'



Key Points

- The error in the sentence is in the segment "**difficult tasks demand lot.**"
- The word "**lot**" should be preceded by the article "**a**" to make the phrase grammatically correct.
- The corrected part of the sentence should be "**difficult tasks demand a lot.**"

Therefore, the **correct sentence** is: "*It is believed that difficult tasks demand a lot of our resources.*"



Additional Information

- **Option 1:** "It is believed that" is grammatically correct.
 - **Option 4:** "of our resources" is grammatically correct.
-

Question 57

Some parts of a sentence have been jumbled up, and labelled P, Q, R and S. Select the option that gives the correct sequence in which these parts can be rearranged to form a meaningful and grammatically correct sentence.

The Government of India Act was passed by

P. separate acts namely, the Government of India Act 1935 and

Q. the longest act enacted by the British Parliament

R. the British Parliament in August 1935, it was

S. at that time, hence it was divided into two

the Government of Burma Act 1935,

Options:

A. QSPR

B. PQRS

C. SPQR

D. RQSP

Answer: D

Solution:

The correct answer is 'RQSP'.



Key Points

- The sentence begins with the main subject "The Government of India Act", which was passed by "the British Parliament in August 1935", indicating the starting point as segment R.
- The next logical information is the fact that it was "the longest act enacted by the British Parliament", which is given in segment Q.
- Due to its length, it was divided into two separate acts, as described in segments S and P.
- Thus, the complete sentence should read: "The Government of India Act was passed by the British Parliament in August 1935, it was the longest act enacted by the British Parliament at that time, hence it was divided into two separate acts namely, the Government of India Act 1935 and the Government of Burma Act 1935."

Therefore, the correct answer is Option 4.

Question 58

A sentence has been given with a blank to be filled with an appropriate option. Choose the correct alternative.

I _____ the cake before my mother arrived.

Options:

A. had baked

B. bake

C. baked

D. have baked

Answer: A

Solution:

The correct answer is 'Option 1' i.e. "had baked."

Key Points

- In this sentence, two past events are mentioned: baking the cake and mother's arrival.
- When **two actions occur** in the **past**, the action that happened first is usually in the past perfect tense, while the action that happened second is in the simple past tense.
- In this sentence, "*before my mother arrived*" is in the **simple past**, so the baking of the cake, which happened first, should be in the past perfect ("had baked").

Therefore, the **correct sentence** is: '*I **had baked** the cake before my mother arrived.*'

Additional Information

- Here's the explanation for each option:
 - "**Bake**" - This is the base form of the verb, typically used in the present tense. It is not suitable for this sentence, which refers to past actions.
 - "**Baked**" - This is the past simple form of the verb. It can be used to indicate a completed action in the past, but when describing two events that both occurred in the past, the action that happened first (baking the cake) should be in the past perfect tense.
 - "**Have baked**" - This is the present perfect form of the verb. It's used to connect the past and the present, often to describe an action that happened in the past but is still relevant in the present. It does not fit in this sentence, which refers to two distinct past events.
-

Question 59

Choose the word opposite in meaning to the given word.

UNSCRUPULOUS

Options:

- A. Dedicated
- B. Single-minded
- C. Superfluous
- D. Conscientious

Answer: D

Solution:

The correct solution is **option 4 i.e. Conscientious**.



Explanation

Let us know the meaning of the words given:

- **Dedicated** means to be devoted to a task or purpose.
- **Single-minded** means concentrating on only one aim.
- **Superfluous** means unnecessary, especially through being more than enough.
- **Conscientious** means wishing to do one's work or duty well and thoroughly.

The given word '**Unscrupulous**' means having or showing no moral principles; not honest or fair. So, the word opposite in meaning to the given word is '**Conscientious**'.

Question 60

Here is a letter to MTNL complaining that your phone has not been working for a few days. The sentences have got jumbled up. Rearrange the sentences to make the complaint coherent.

I. My landline phone has not been working for a week.

II. I am a resident of House No. X, Street Y, Chennai.

III. Please set my phone right at the earliest.

IV. I informed the Area Manager telephonically about it but there is no response.

Options:

A. I, II, III, IV

B. II, I, IV, III

C. IV, III, II, I

D. III, II, IV, I

Answer: B

Solution:

The correct answer is 'II, I, IV, III'.

Key Points

- The **first** sentence will introduce the sender and his/her address. Hence, '**II**' will be the first sentence of the letter.
- The **second** sentence will explain the problem. Hence, '**I**' will be the second sentence of the letter.
- The next sentence will be '**IV**' as it states the sender's efforts of contacting the **Area Manager**.
- The last statement will be **III**, which is **requesting** to repair the phone as early as possible.

Hence, the correct answer is **Option 2**.

Question 61

**Four words are given out of which only one word is correctly spelt.
Find the correctly spelt word.**

Options:

- A. Greatfull
- B. Greatful
- C. Gratefull
- D. Grateful

Answer: D

Solution:

The correct answer is "**Grateful**".

Key Points

- "**Grateful**" is the correct spelling of the word.
- **Grateful** - It means **feeling or showing appreciation for something received or done**.
- When you are grateful, you are thankful for the kindness, help, or opportunities that you have received.
 - **For Example:** "I am truly **grateful** for all the support you've given me during my difficult times."

Therefore, the correct answer is "**Option 4**".

Question 62

Direction: Select the most appropriate option to fill in the blank.

I enjoy finding ways to make the most of my budget by being _____.

Options:

A. lethargic

B. miser

C. frugal

D. gloomy

Answer: C

Solution:

The correct answer is 'Option 3' i.e. 'Frugal'.



Key Points

- The **sentence is about** maximizing the use of one's budget.
- The **blank should be filled** with a word that denotes careful or sparing use of money.
- "**Frugal**" (मितव्ययी) means economical in use or expenditure; not wasteful.

Therefore, the **correct sentence** is: "*I enjoy finding ways to make the most of my budget by being frugal.*"



Additional Information

- "**Lethargic**" (आलसी) means sluggish and lacking in energy, and does not relate to the context.
 - "**Miser**" (कंजूस) refers to a person who hoards money and avoids spending, but this term can have a negative connotation.
 - "**Gloomy**" (उदास) means dark or dim; deeply shaded, and is not related to the context of the sentence.
-

Question 63

Choose the option that is the active form of the sentence.

The roads have been dug up again by the municipal authorities.

Options:

A. The municipal authorities have dug up the roads again.

B. The municipal authorities are being dug up the roads again.

C. The municipal authorities are dug up the roads again.

D. The municipal authorities were dug up the roads again.

Answer: A

Solution:

The correct answer is "The municipal authorities have dug up the roads again".



Key Points

- In order to convert the given sentence into an active form, we need to identify the **tense** first of all.
- The given sentence is in the **present perfect tense**.
- The rule of the present perfect tense (active voice)-**Has/Have + Verb 3rd form**.
- The subject is '**the municipal authorities**' and the object is '**the roads**'.
- So the sentence would be -*The municipal authorities have dug up the roads again.*

Hence, the correct answer is **Option 1**.



Additional Information

- The criminal has been punished by the jailer.
 - *The jailer has punished the criminal.*
- The clothes have been washed by my friend.
 - *My friend has washed the clothes.*



Hinglish

- दिए गए वाक्य को सक्रिय रूप में बदलने के लिए, हमें सबसे पहले **tense** की पहचान करनी होगी।
- दिया गया वाक्य **present perfect tense** में है।
- **Present perfect tense** (active voice) का नियम **Has/Have + Verb 3rd form** है।
- The subject is '**the municipal authorities**' तथा Object is '**the roads**'.
- तो वाक्य होगा -*The municipal authorities have dug up the roads again.*

अतः सही उत्तर **विकल्प 1** है।

Question 64

Which of the following is the correct noun form of the word 'Destroy' ?

Options:

- A. Destructive
- B. Destruction
- C. Destruct
- D. Destructing

Answer: B

Solution:

The correct answer is '**Option 2**', destruction.



Key Points

- The correct noun form of the word 'Destroy' is Destruction.
- Destructive is an adjective.
- Destruct is not commonly used as a noun.
- Destructing is a present participle form of the verb.

Hence, the correct answer is **option 2**.

Correct sentence: *The correct noun form of the word 'Destroy' is Destruction.*



Additional Information

- The noun 'Destruction' refers to the action or process of causing so much damage to something that it no longer exists or cannot be repaired.
- Examples of usage:
 - The hurricane caused widespread **destruction**.
 - The war led to the **destruction** of many historic buildings.

Question 65

Select the option that expresses the given sentence in passive voice.

His parents will send him to a boarding school next year

Options:

- A. He will be sent to a boarding school next year by his parents.
- B. He will be sent to a boarding school by his parents next year.

C. He is to be sent to a boarding school next year by his parents.

D. He would be sent to a boarding school next year by his parents.

Answer: A

Solution:

The correct answer is: "**He will be sent to a boarding school next year by his parents.**"



Key Points

- In passive voice, the subject of the active sentence becomes the receiver of the action, and the agent (the doer of the action) is either mentioned after the verb with the preposition "by" or omitted if it's not important to the sentence's meaning.
- The given sentence is in active voice with the subject "**His parents**" performing the action of sending "**him**" to a boarding school.
- To convert it into passive voice, we need to make the object "**him**" the subject of the sentence and add the helping verb "**will be**" followed by the past participle of the main verb "**sent**".
- In the passive sentence, "**He will be sent**" is the passive construction of "**His parents will send,**" and "**by his parents**" indicates who is performing the action in a passive voice sentence.
- The preposition "**by**" is used to indicate the doer of the action, which is "his parents" in this case.

Therefore, the correct answer is **option 1**.

Question 66

Select the most appropriate synonym of the given word.

Grandeur

Options:

A. Magnificence

B. Liable

C. Commotion

D. Power

Answer: A

Solution:

The correct answer is 'Option 1' i.e. 'Magnificence'.

Key Points

- The word '**Grandeur**' typically means the quality of being grand or magnificent. (भव्यता).
 - **Example:** The grandeur of the royal palace left all the visitors in awe.
- The synonym '**Magnificence**' signifies the quality of being impressively beautiful or grand.(अति उत्कृष्टता).
 - **Example:** The magnificence of the natural landscapes is truly breathtaking.

Therefore, the **correct answer** is '**Magnificence**'.

Additional Information

- '**Liabile**' as per Option 2 is an adjective that means responsible by law or likely to do or to be something. (उत्तरदायी).
 - '**Commotion**' as per Option 3 is a noun that suggests a state of confused and noisy disturbance.(हलचल).
 - '**Power**' as per Option 4 typically refers to the ability to do something or act in a particular way, or the capacity to influence the behavior of others.(शक्ति).
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Question 67

Read the passage given below and answer the questions that follow.

Almond trees are deciduous with a hardy dormancy. Typically growing 3–4.5 metres (10–15 feet) tall, the trees are strikingly beautiful when in flower; they produce fragrant, five-petaled, light pink to white flowers from late January to early April north of the Equator. The flowers are self-incompatible and thus require insect pollinators to facilitate cross-pollination with other cultivars. The growing fruit (a drupe) resembles a peach until it approaches maturity; as it ripens, the leathery outer covering, or hull, splits open, curls outward, and discharges the pit. Despite their common label, almonds are not true nuts (a type of dry fruit) but rather seeds enclosed in a hard fruit covering.

The sweet almond is cultivated extensively in certain favourable regions, though nut crops are uncertain wherever frosts are likely to occur during flowering. While more than 25 types of almonds are grown in California, Marcona and Valencia almonds come from Spain, and Ferragnes are imported from Greece.

In which months, does the Almond tree produce flower?

Options:

- A. Late January to early April
- B. Late April to early January
- C. January
- D. March to May

Answer: A

Solution:

The correct answer is "**Late January to early April**"

Key Points

- You can find the highlighted lines in the **1st** paragraph:
 - Almond trees are deciduous with a hardy dormancy. Typically growing 3–4.5 metres (10–15 feet) tall, the trees are strikingly beautiful when in flower; they produce fragrant, five-petaled, light pink to white flowers from **late January to early April north** of the Equator. The flowers are self-incompatible and thus require insect pollinators to facilitate cross-pollination with other cultivars. The growing fruit (a drupe) resembles a peach until it approaches maturity; as it ripens, the leathery outer covering, or hull, splits open, curls outward, and discharges the pit. Despite their common label, almonds are not true nuts (a type of dry fruit) but rather seeds enclosed in a hard fruit covering.

Hence, the correct answer: *Late January to early April*

Question 68

Read the passage given below and answer the questions that follow.

Almond trees are deciduous with a hardy dormancy. Typically growing 3–4.5 metres (10–15 feet) tall, the trees are strikingly beautiful when in flower; they produce fragrant, five-petaled, light pink to white flowers from late January to early April north of the Equator. The flowers are self-incompatible and thus require insect pollinators to facilitate cross-pollination with other cultivars. The growing fruit (a drupe) resembles a peach until it approaches maturity; as it ripens, the leathery outer covering, or hull, splits open, curls outward, and discharges the pit. Despite their common label, almonds are not true nuts (a type of dry fruit) but rather seeds enclosed in a hard fruit covering.

The sweet almond is cultivated extensively in certain favourable regions, though nut crops are uncertain wherever frosts are likely to occur during flowering. While more than 25 types of almonds are grown in California, Marcona and Valencia almonds come from Spain, and Ferragnes are imported from Greece.

Which of the following options is correct?

Options:

- A. The flowers of Almond have five petals.
- B. The flowers are self-incompatible
- C. The flowers require insect pollinators to facilitate cross-pollination
- D. All of the above

Answer: D

Solution:

The correct answer is "All of the above"

Key Points

- You can find the highlighted lines in the **1st** paragraph:
 - Almond trees are deciduous with a hardy dormancy. Typically growing 3–4.5 metres (10–15 feet) tall, the trees are strikingly beautiful when in flower; they produce fragrant, **five-petaled**, light pink to white flowers from late January to early April north of the Equator. **The flowers are self-incompatible** and thus **require insect pollinators to facilitate cross-pollination with other cultivars**. The growing fruit (a drupe) resembles a peach until it approaches maturity; as it ripens, the leathery outer covering, or hull, splits open, curls outward, and discharges the pit. Despite their common label, almonds are not true nuts (a type of dry fruit) but rather seeds enclosed in a hard fruit covering.

Hence, the correct answer: *All of the above*

Question 69

Read the passage given below and answer the questions that follow.

Almond trees are deciduous with a hardy dormancy. Typically growing 3–4.5 metres (10–15 feet) tall, the trees are strikingly beautiful when in flower; they produce fragrant, five-petaled, light pink to white flowers from late January to early April north of the Equator. The flowers are self-incompatible and thus require insect pollinators to facilitate cross-pollination with other cultivars. The growing fruit (a drupe) resembles a peach until it approaches maturity; as it ripens, the leathery outer covering, or hull, splits open, curls outward, and discharges the pit. Despite their common label, almonds are not true nuts (a type of dry fruit) but rather seeds enclosed in a hard fruit covering.

The sweet almond is cultivated extensively in certain favourable regions, though nut crops are uncertain wherever frosts are likely to occur during flowering. While more than 25 types of almonds are grown in California, Marcona and Valencia almonds come from Spain, and Ferragnes are imported from Greece.

What can be deduced from the given passage?

Options:

- A. Almond trees absorb a lot of water
- B. Almonds are not true nuts
- C. Almonds are exported from India to America
- D. None of the above

Answer: B

Solution:

The correct answer is "**Almonds are not true nuts**"



Key Points

- You can find the highlighted lines in the **1st** paragraph:
 - Almond trees are deciduous with a hardy dormancy. Typically growing 3–4.5 metres (10–15 feet) tall, the trees are strikingly beautiful when in flower; they produce fragrant, five-petaled, light pink to white flowers from late January to early April north of the Equator. The flowers are self-incompatible and thus require insect pollinators to facilitate cross-pollination with other cultivars. The growing fruit (a drupe) resembles a peach until it approaches maturity; as it ripens, the leathery outer covering, or hull, splits open, curls outward, and discharges the pit. Despite their common label, **almonds are not true nuts (a type of dry fruit) but rather seeds enclosed in a hard fruit covering.**

Hence, the correct answer: *Almonds are not true nuts*

Question 70

Read the passage given below and answer the questions that follow.

Almond trees are deciduous with a hardy dormancy. Typically growing 3–4.5 metres (10–15 feet) tall, the trees are strikingly beautiful when in flower; they produce fragrant, five-petaled, light pink to white flowers from late January to early April north of the Equator. The flowers are self-incompatible and thus require insect pollinators to facilitate cross-pollination with other cultivars. The growing fruit (a drupe) resembles a peach until it approaches maturity; as it ripens, the leathery outer covering, or hull, splits open, curls outward, and discharges the pit. Despite their common label, almonds are not true nuts (a type of dry fruit) but rather seeds enclosed in a hard fruit covering.

The sweet almond is cultivated extensively in certain favourable regions, though nut crops are uncertain wherever frosts are likely to occur during flowering. While more than 25 types of almonds are grown in California, Marcona and Valencia almonds come from Spain, and Ferragnes are imported from Greece.

What does the word 'extensively' mean in the given passage?

Options:

- A. in a way that covers or affects a large area.
- B. in a way that covers or affects a tiny area.
- C. found in a particular region
- D. covering a sandy area

Answer: A

Solution:

The correct answer is "**in a way that covers or affects a large area.**"



Key Points

- Let's look at the meaning of the given word:
 - **Extensively**-in a way that covers or affects a large area.
- Look at the example:
 - The house was **extensively** damaged by the fire

Hence, the correct answer: *in a way that covers or affects a large area*



Hinglish

- **Extensively**-बगीचे का एक ऐसा क्षेत्र जहाँ पर छोटे छोटे पौधे लगे होते हैं
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